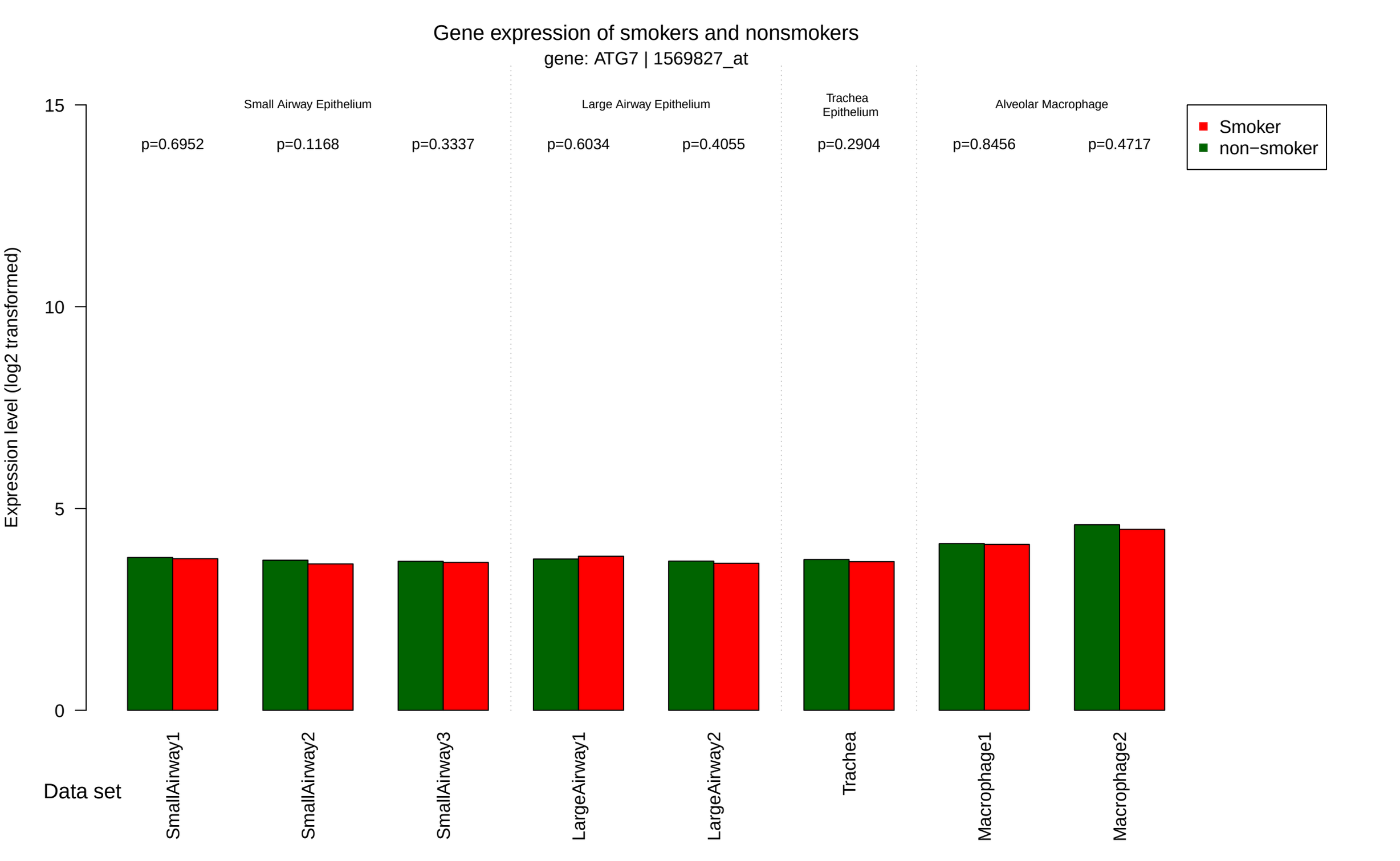
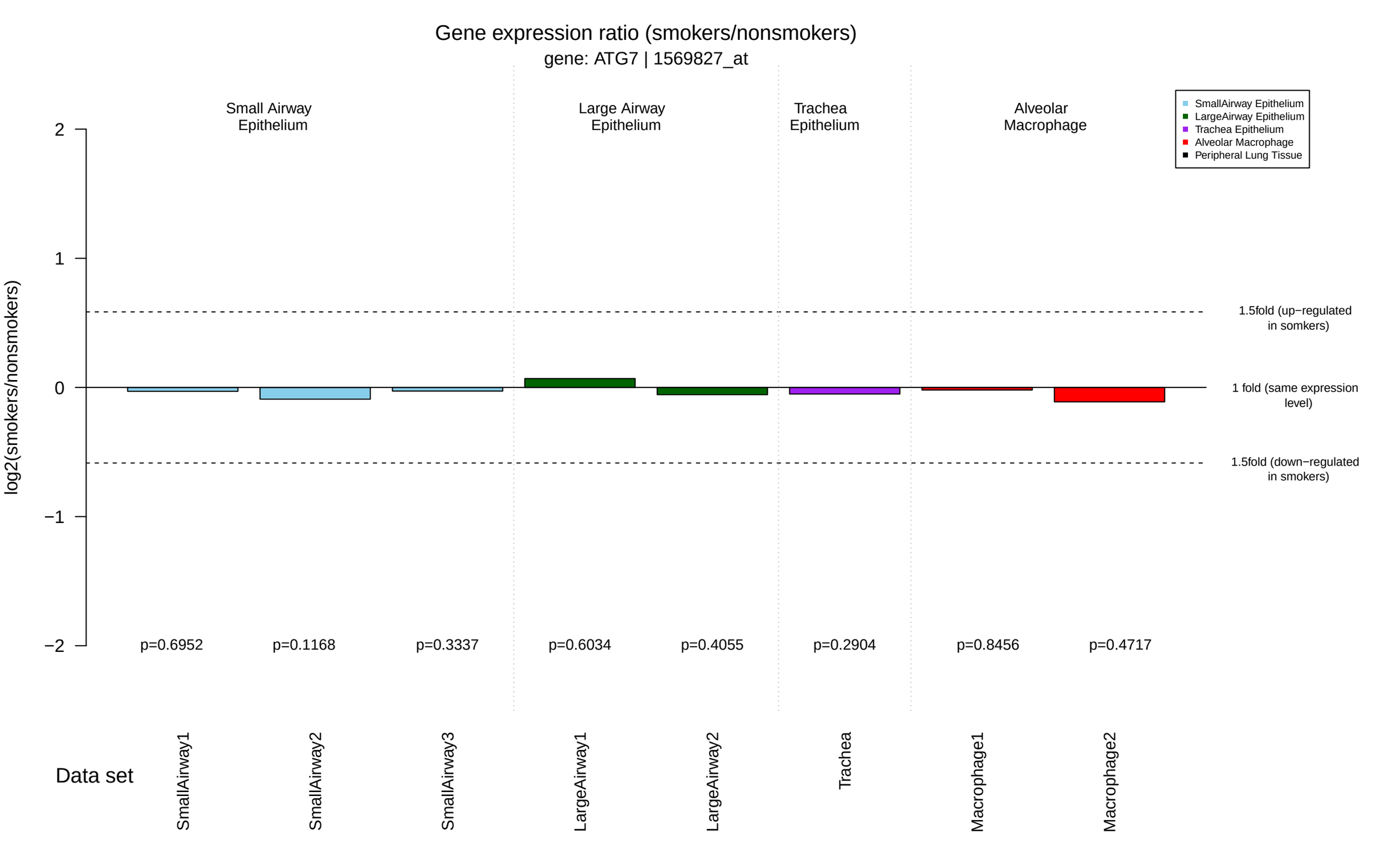
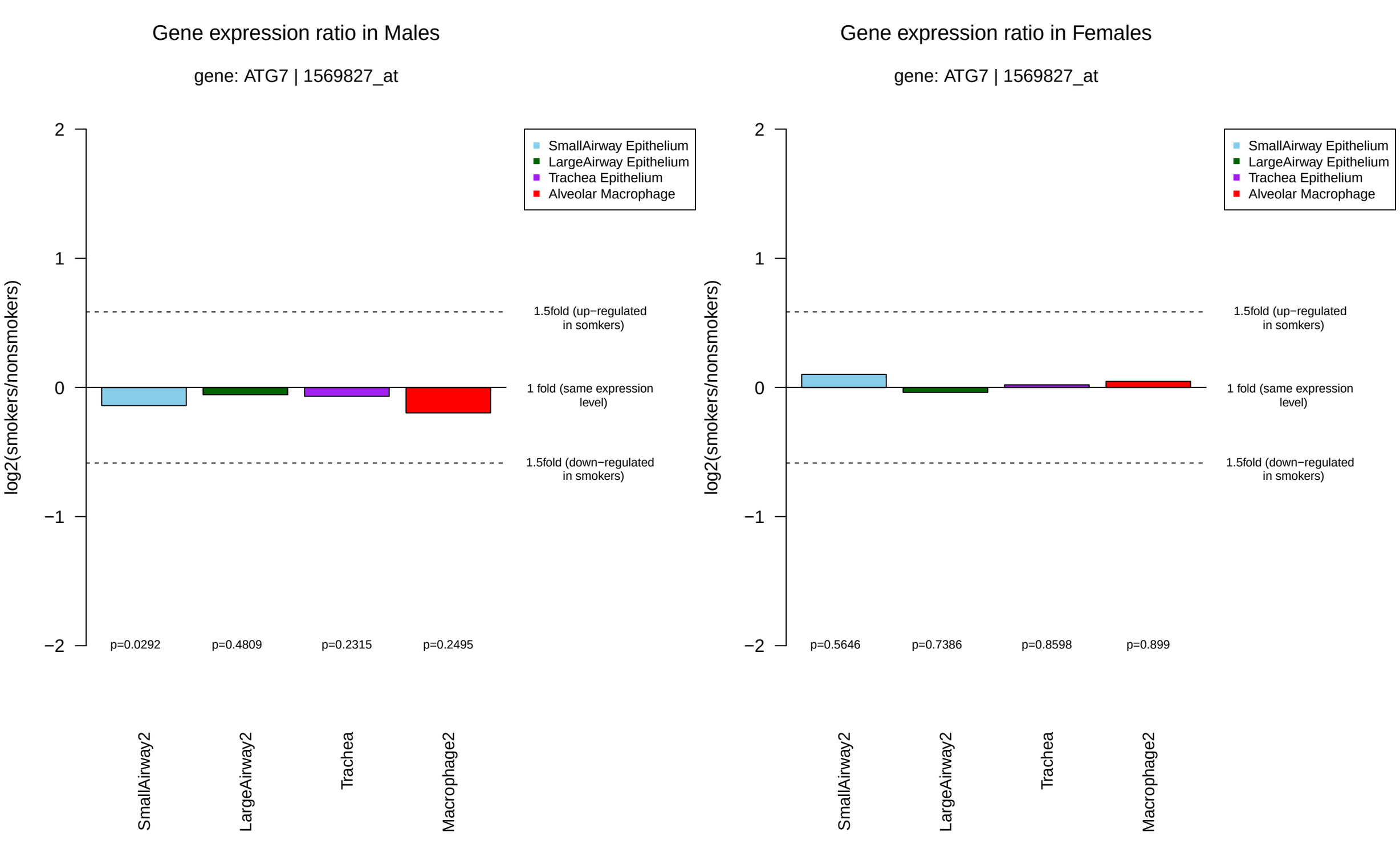
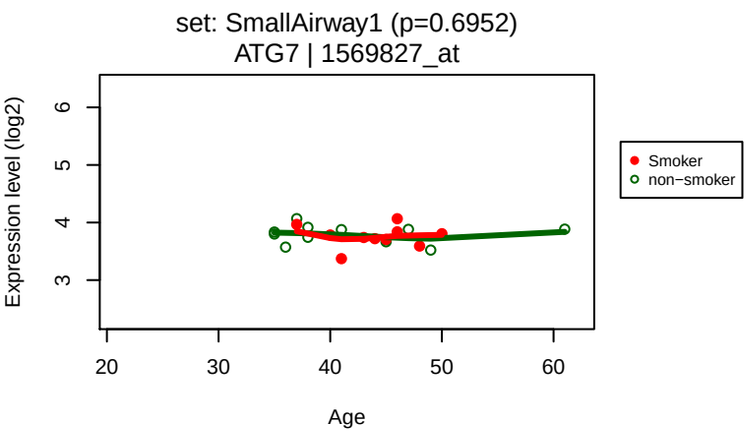
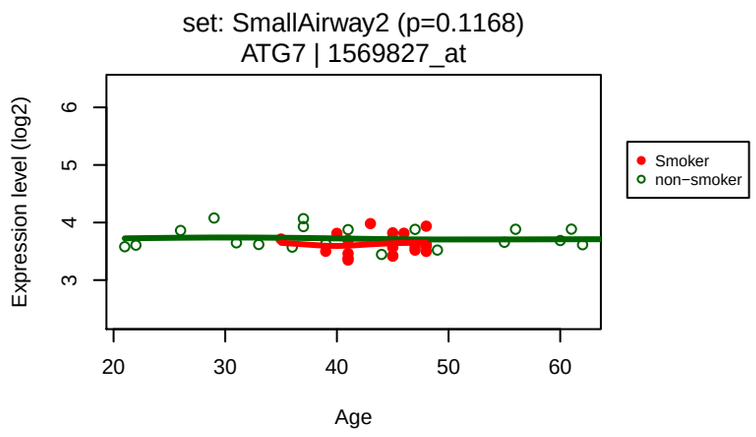
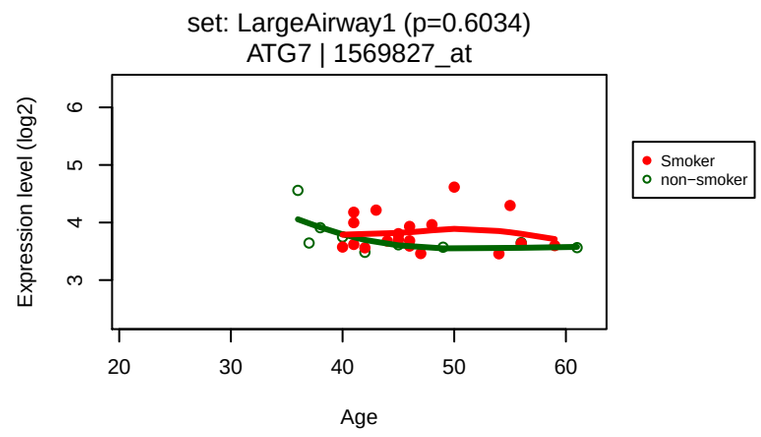

Gene expression of smokers and nonsmokers
gene: ATG7 | 1569827_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6952
p=0.1168
p=0.3337
p=0.6034
p=0.4055
p=0.2904
p=0.8456
p=0.4717
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

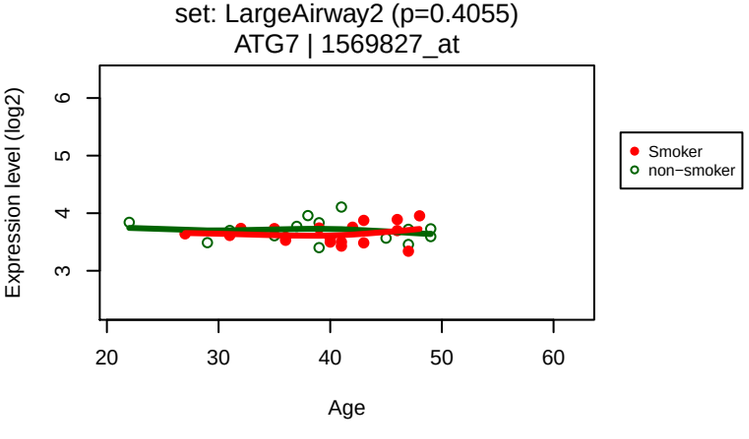

Gene expression ratio (smokers/nonsmokers)
gene: ATG7 | 1569827_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6952
p=0.1168
p=0.3337
p=0.6034
p=0.4055
p=0.2904
p=0.8456
p=0.4717
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

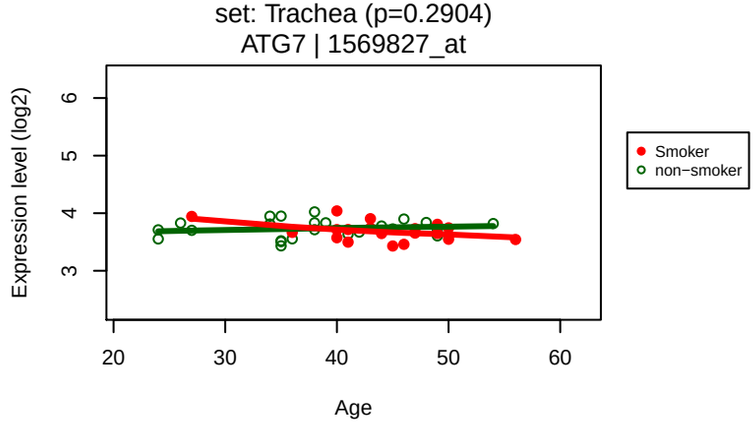

Gene expression ratio in Males
gene: ATG7 | 1569827_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0292
p=0.4809
p=0.2315
p=0.2495
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATG7 | 1569827_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5646
p=0.7386
p=0.8598
p=0.899
SmallAirway2
LargeAirway2
Trachea
Macrophage2

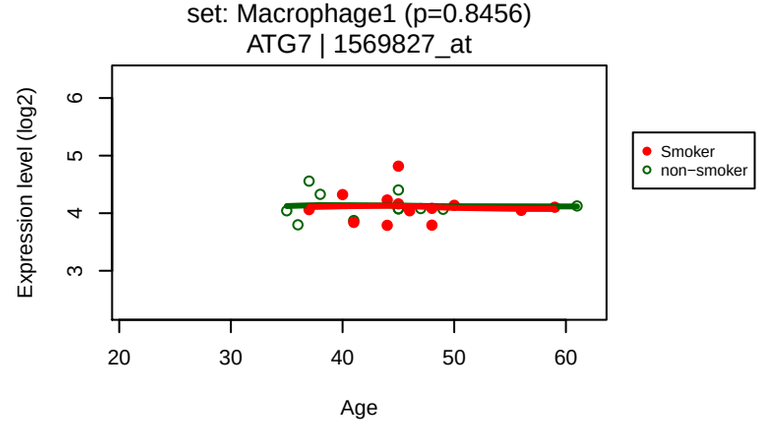

set: SmallAirway1 (p=0.6952)
ATG7 | 1569827_at
Expression level (log2)
Age
Smoker
non−smoker

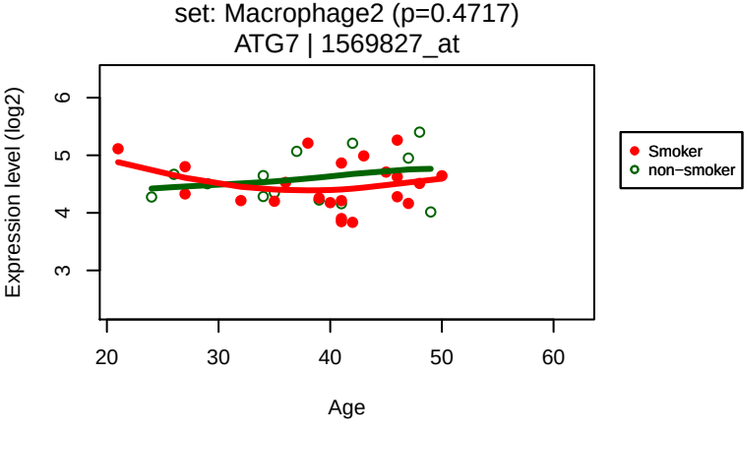

set: SmallAirway2 (p=0.1168)
ATG7 | 1569827_at
Expression level (log2)
Age
Smoker
non−smoker

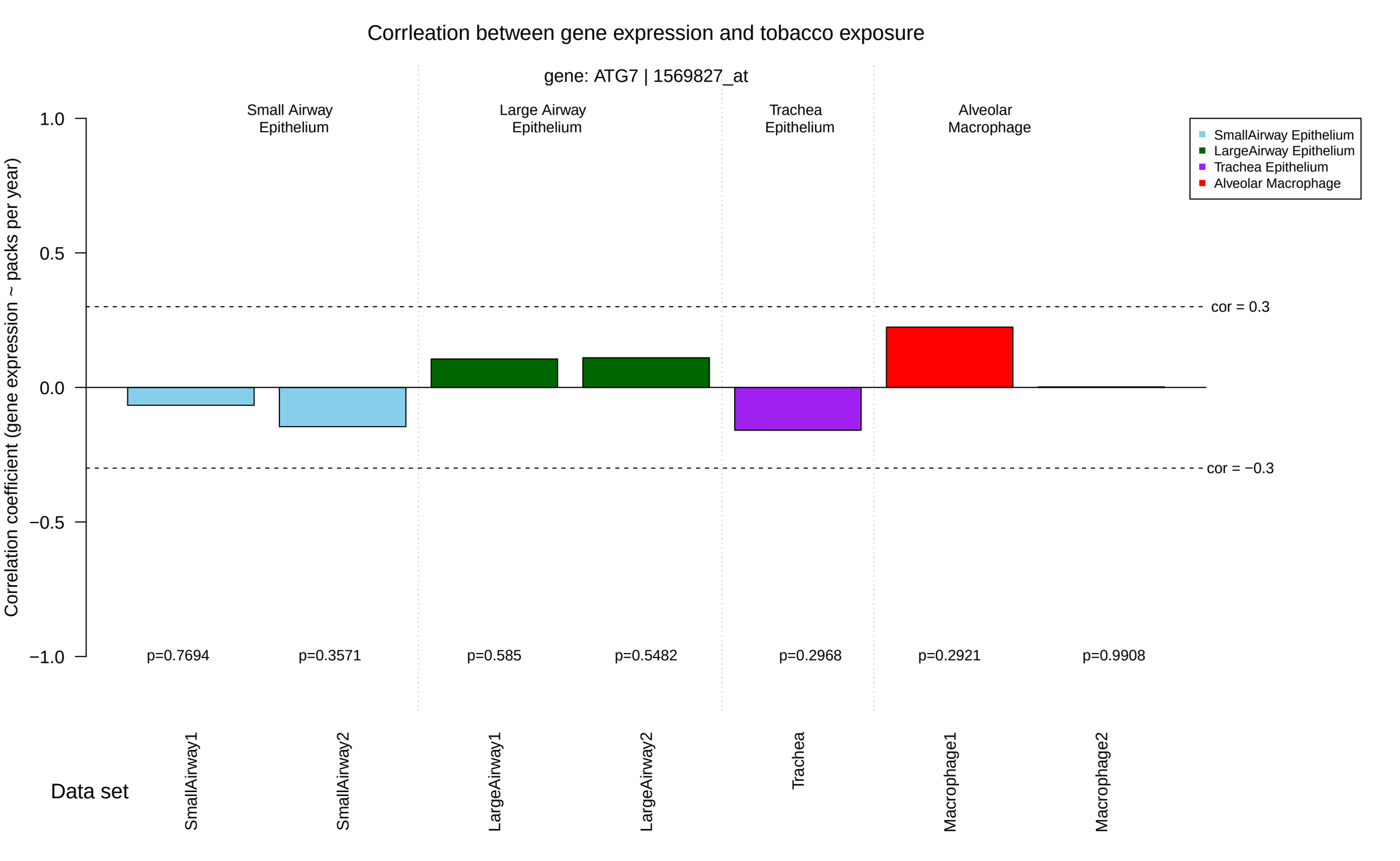

set: LargeAirway1 (p=0.6034)
ATG7 | 1569827_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4055)
ATG7 | 1569827_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2904)
ATG7 | 1569827_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8456)
ATG7 | 1569827_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4717)
ATG7 | 1569827_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATG7 | 1569827_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7694
p=0.3571
p=0.585
p=0.5482
p=0.2968
p=0.2921
p=0.9908
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2